



CHAPTER

Chapter 5 Data Visualization

Data Analytics Using Python
Dr. Ziping Liu

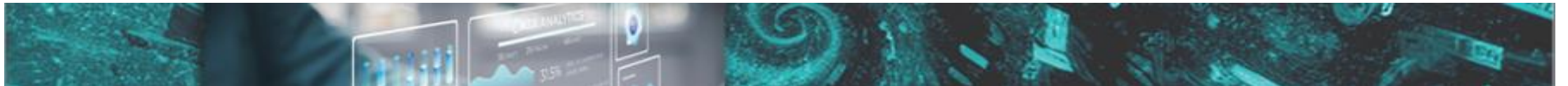


Chapter 5 Python Programming Language Review

<https://he.kendallhunt.com/product/data-analytics-using-python>

Data Visualization

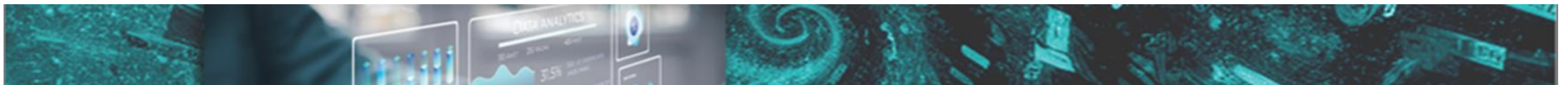
- Visual representations include graphs and charts
 - can more clearly reveal patterns, trends, correlations, and anomalies that might be difficult to discern from raw data alone.
- Common types of visualizations include
 - line charts, histograms, box plots, scatter plots, and bar charts.
- Python-based plotting libraries
 - matplotlib, seaborn, pandas
 - create visualizations that facilitate trend analysis, distribution examination, correlation exploration, and comparison



Plotting in Pandas

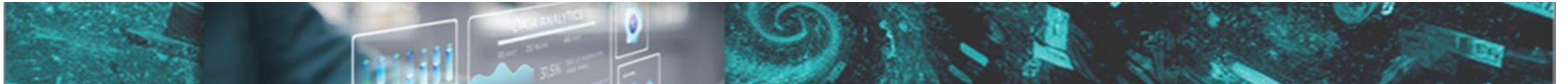
- Pandas offers quick and simple data visualization for DataFrames and Series
- methods to do various plotting

Plotting types	Purposes
Line	Trend for continuous data
Bar	Comparison on different categories
Histogram or box	Data distribution
Scatter	Data correlations, patterns, or clusters
Area	Accumulation of each data category
Barh	Horizontal bar plot
Pie	Data proportion
KDE(Kernel Density Estimation)	Likelihood of encountering specific values in a range



Plotting in Matplotlib

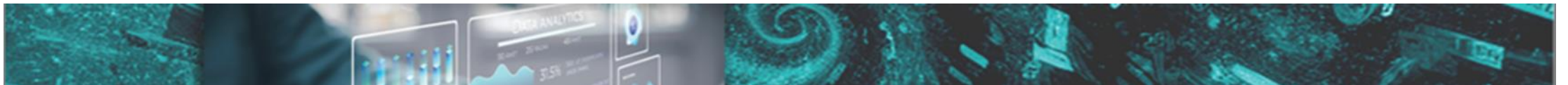
- Pandas lacks the options to customize the plotting canvas and other aspects of the plot, such as line styles and colors.
- Matplotlib offers more complex and informative plotting for data exploratory
- Matplotlib is a Python package for 2D plotting that offers an object-oriented approach to create a plotting canvas and arrange various plot elements on it



Plotting in Matplotlib

- Installation and Testing

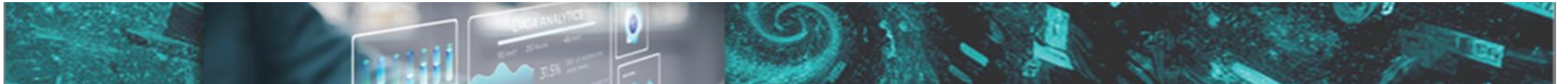
Using pip:	Using conda:
<code>pip install matplotlib</code>	<code>conda install matplotlib</code>
Verifying the installation:	Verifying the installation:
<code>pip show matplotlib</code>	<code>conda list matplotlib</code>
Testing the Matplotlib installation using a simple script:	
<pre>import matplotlib.pyplot as plt x = [1, 2, 3] y = [7, 8, 9] plt.plot(x, y) plt.show()</pre>	



Plotting in Matplotlib

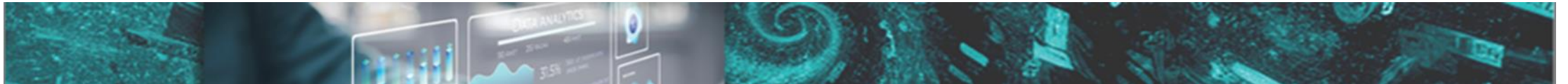
- Getting Started with matplotlib
 - submodule, pyplot provides a collection of functions to create plots and customize their features

```
import matplotlib.pyplot as plt
#calculate x, y coordinate values
x = ...
y = ...
plt.plot(x, y)
plt.xlabel('X-axis Label')
plt.ylabel('Y-axis Label')
plt.title('XXX Plot:')
plt.show()
```



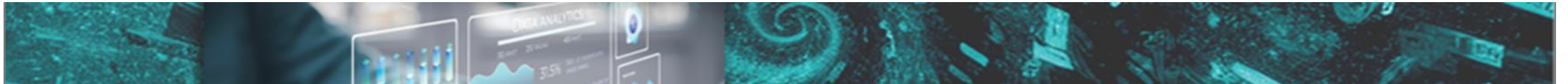
Plotting in Matplotlib

- Getting Started with matplotlib
 - can provide a configurable canvas
 - the figure() function returns a Figure object
 - add subplots to canvas using the add_subplot() function
 - various types of plots can be placed on subplots



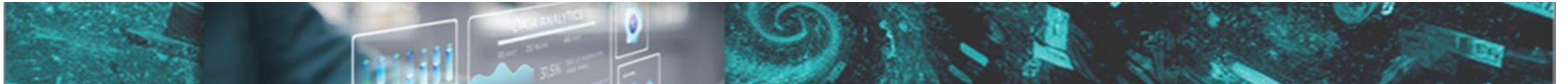
Plotting in Matplotlib

- More on matplotlib
 - Each subplot offers to provide data display through various types of plotting, labels for x and y axis, title for the subplot, grid lines on the subplot, legend, annotation, etc
 - .set_title()
 - .set_xlabel()
 - .set_ylabel()
 - .legend()
 - .annotate()



Plotting in Matplotlib

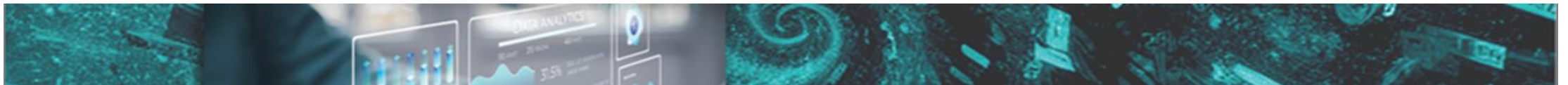
- Integrate `plot()` method from Pandas and plotting panel configuration from matplotlib
 - Pandas `plot()`
 - provides an argument to specify plotting on a specific subplot created by matplotlib's `subplots()` method or `add_subplot()` method
 - allows multiple plots shown on the same display panel in the case that Pandas's `plot()` is chosen



Plotting in Matplotlib

- Integrate plot() method from Pandas and plotting panel configuration from matplotlib

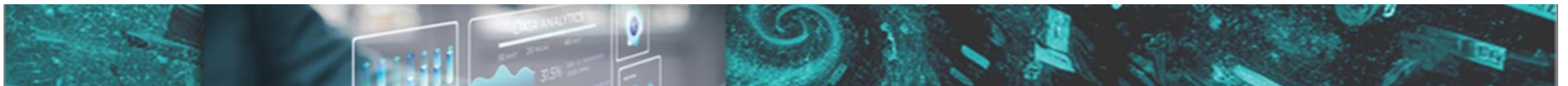
Pandas plot() method	matplotlib's plotting methods
belongs to DataFrame;	doesn't understand DataFrame;
understands DataFrame structure and automatically extracts data for plotting based on column names	plotting needs x and y axis coordinate values from user input
offers a basic and quick approach for exploratory data analysis (EDA) via visualization	Offers more complex and customized features for plots



Plotting in Seaborn

- Seaborn provides the high-level interface to simplify the statistical visualization analysis of Pandas DataFrame.
- Common functions for plotting include violin plots, box plots, joint plots, pair plots, and other statistical visualizations.

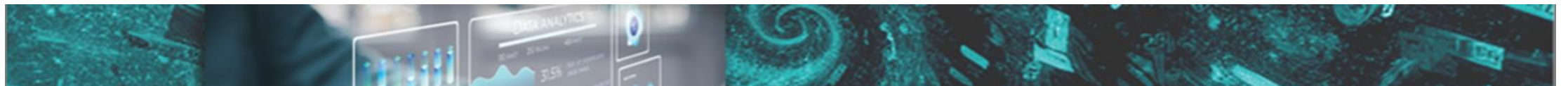
seaborn	matplotlib
statistical data visualization	general-purpose plotting
less customizable; good for basic data visualization with readability and aesthetics	highly flexible and customizable; offers more complex features
coding is more concise	coding is more verbose



Plotting in Seaborn

- Installation and Testing

Using pip:	Using conda:
<code>pip install seaborn</code>	<code>conda install seaborn</code>
Verifying the installation:	Verifying the installation:
<code>pip show seaborn</code>	<code>conda list seaborn</code>
Testing the seaborn installation using a simple script:	
<pre>import seaborn as sns # Create iris dataframe directly from the Iris dataset iris = sns.load_dataset("iris") print(iris.head()) # plot sepal_length distribution sns.displot(iris["sepal_length"])</pre>	



Plotting in Seaborn

- Working with seaborn
 - To plot with seaborn is similar as to plot in matplotlib, call the respective plot method, provide the data and/or features
 - .lineplot(data, x, y, hue)
 - .barplot(data, y, x, hue, orient)
 - .boxplot(data, x, y, orient)
 - .pairplot(data)
 - .regplot(x, y, data)
 - .catplot(x, y, hue, col, data, kind, height, aspect)
 - .jointplot(x, y, data, kind)
 - .violinplot(x, y, data, inner)
 - For the configurations of the plotting panel and plot settings, we can use matplotlib

